

Music Diary: Using ChatGPT to Craft Interactive Diaries with Emotional Music for Reflection and Sharing Real-life Emotions

Jui-Wei Huang
National Tsing Hua University
Hsinchu, Taiwan
ahhhhuang@gapp.nthu.edu.tw

Yan-Ming Chen
National Tsing Hua University
Hsinchu, Taiwan
gepz@gapp.nthu.edu.tw

Chuang-Wen You
National Tsing Hua University
Hsinchu, Taiwan
cwyou@mx.nthu.edu.tw



Figure 1: A promotional image of the Music Diary shows how people can easily and comfortably use the app. With Music Diary, users can enjoy recording their lives while enhancing their emotional awareness through music.

ABSTRACT

A daily diary can benefit one's mental health and well-being by providing an outlet for emotional expression, self-reflection, and stress reduction. However, the traditional approach to diary writing is hindered by the lack of an intuitive mechanism to deal with emotional experiences via textual descriptions and a gradual decline in writing skills with the adoption of social media and messaging apps. This paper introduces a Music Diary app, which uses ChatGPT to summarize text and generate music that echoes the emotional state of the user. The results of this pilot study revealed that the Music Diary app potentially facilitates the creation of diary entries, while the musical soundtrack promotes intuitive awareness, reflection, and expression as well as one's willingness to share experiences on social platforms. In experiments, the proposed system was shown to make the process of keeping a diary more enjoyable and the novel format makes it possible to expand this personal activity to a wider social realm.

CCS CONCEPTS

• Human-centered computing → Ubiquitous and mobile computing systems and tools; User interface toolkits.

KEYWORDS

Diary, Chatbot, Music, Emotional Expression, Retrospection, Self-awareness, Emotional Social Sharing

ACM Reference Format:

Jui-Wei Huang, Yan-Ming Chen, and Chuang-Wen You. 2024. Music Diary: Using ChatGPT to Craft Interactive Diaries with Emotional Music for Reflection and Sharing Real-life Emotions. In *Companion of the 2024 ACM International Joint Conference on Pervasive and Ubiquitous Computing Pervasive and Ubiquitous Computing (UbiComp Companion '24)*, October 5–9, 2024, Melbourne, VIC, Australia. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3675094.3677550>

1 INTRODUCTION

The process of keeping a daily diary or journal can benefit one's well-being by providing a tool for self-reflection and an outlet for emotional expression [5]. It provides the opportunity to revisit experiences, ponder one's feelings, and gain a deeper self-awareness [11]. Moreover, the advent of social media and messaging apps has promoted the use of fragmented text, which does not provide the depth required for a contemplative assessment of emotional issues or the rigor required to put events into context and finish diary entries. Most existing diary apps rely on text inputs, which require repeated readings to discern underlying emotional content. Moreover, the format of these apps prioritizes privacy rather than the sharing of one's thoughts with others [12], which is crucial to a sense of satisfaction and well-being [10].

A number of apps, such as Grid Diary[8], have been developed to deal with these issues through the provision of prompts to encourage the writing of related stories. Grid Diary [8] scatters common prompts in a novel user interface (UI) comprising a three-by-three grid of squares. Journal [6] selects emotive photographs as prompts. Nonetheless, the inability of these apps to engage in continuous

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

UbiComp Companion '24, October 5–9, 2024, Melbourne, VIC, Australia

© 2024 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1058-2/24/10

<https://doi.org/10.1145/3675094.3677550>

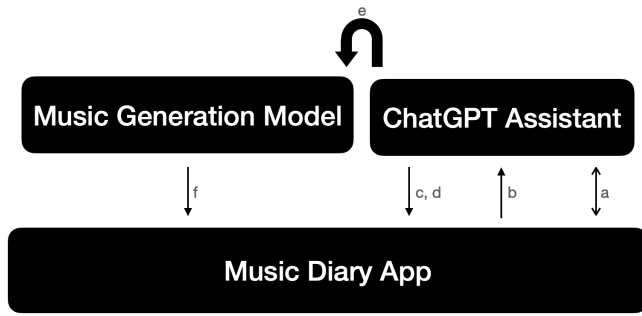


Figure 2: The interaction between three main components: Music Diary App, ChatGPT Assistant, and Music Generation Model. (a) Users will engage in real-time text conversations with the ChatGPT assistant through the music diary app. (b) Conversation records from the interactions will be saved and uploaded to the ChatGPT assistant for summarization. (c) Users will receive the summarized diary text content. (d) Also, they will also receive an analysis of the main emotions. (e) The main emotions will be used as keywords to request emotional music from the music generation model. (f) The generated emotional music will be downloaded and integrated with the diary text content and main emotions to create a complete music diary for the user.

interactive conversations precludes the exploration of profound feelings or meaningful interactions. Other apps focus on emotional expression and retrospection. Clarity [4] and Mojito [7] utilize emojis to facilitate emotional expression; however, the fixed format of emojis limits the range of emotions that can be expressed, while the predefined emotions they portray provide only a superficial rendering of the complex abstractions that make up the human psyche. Some apps, such as Journify [2], are meant to facilitate the process of keeping a journal. Journify [2] uses vocal inputs as a diary medium and encourages content sharing with friends. Nonetheless, vocal expression lacks the review mechanism required to ponder daily events, and privacy concerns can reduce the willingness to share.

The tracks in the album "12" by Ryuichi Sakamoto [13] are named after dates corresponding to the thoughts and reflections experienced on specific days. This concept inspired our use of music as a medium to document daily life events. In this paper, we proposed the Music Diary system (as shown in Figure 2) which employs a chatbot (powered by ChatGPT) to summarize text describing daily-life events and identify the corresponding emotions. We also employed a music generation model (powered by Audiogen [3]) to create music that is representative of the identified emotions. The chatbot guides users a question-and-answer format to recollect daily events and feelings and then integrates those insights with chat logs to complete the diary. The ability of ChatGPT to detect emotional content in text [9], makes it possible for the system to assign an emotional fingerprint representative of a given day. The emotional fingerprint is then used to guide the generation of a unique representative soundtrack. Essentially, Music Diary composes a soundtrack to accompany a textual summary of daily life events.

This study makes a number of contributions to the literature. First, the proposed system provides a question-and-answer mechanism to facilitate the summary and organization of one's thoughts in the form of a daily diary. The system assigns an emotion fingerprint to those events based on chatbot interactions, and then uses the emotional content to guide the generation of soundtrack aimed at enriching introspective depth and retrospective awareness. It appears that many individuals would be inclined to share their diary entries (text summaries and corresponding emotion-infused music) on social networks. It also appears that the sharing of content can further enhance motivation to keep a diary. The results of a pilot study revealed that ChatGPT can simplify the process of writing a diary, while the accompanying emotional music facilitates introspection and promotes one's willingness to share experiences with others.

2 MUSIC DIARY

Our objectives in developing the Music Diary system were two-fold: 1) Facilitate the creation of music diaries, and 2) facilitate the review of previous music diaries. As shown in Figure 2, the proposed system comprises three core components: Music Diary App, ChatGPT Assistant, and Music Generation Model. We also developed a back-end server to connect the music diary app with various services and enable the sharing of diary entries; however, that the details of server-side operations are beyond the scope of this paper. The interactive process of creating a music diary can be divided into three stages: **Guiding**, **Creating**, and **Reviewing**. As indicated by Figure 2, the Guiding stage involves a real-time Q&A with ChatGPT, where users respond to seven or eight short questions related to daily events (label "a" in Figure 2). The corresponding dialogue records, including all the ChatGPT questions and user responses are buffered locally on the user's device. In the Creating stage, the system uploads the dialogue records to the server (label "b" in Figure 2), after which the ChatGPT assistant returns a summary of diary text content and corresponding emotions (label "c" and "d" in Figure 2). The representative emotions serve as keywords to guide the process of music generation (label "e" in Figure 2). Finally, the system downloads the generated music (label "f" in Figure 2), and combines the diary text content in the form of daily diary entries to be displayed on the user interface of the Music Diary.

2.1 Music Diary User Interface

The diary entries are presented in a calendar view to ensure a clean interface and user-friendly operations. Note that the status of music diaries for each day (in calendar view) are indicated according to color: transparent (not started - in Figure 4(b)), yellow (today's date - in Figure 4(c)), red (unfinished - in Figure 4(d)), and green (completed - in Figure 4(e)). The process of logging a diary entry for the current day involves clicking on a date or the music note icon (in Figure 4(a)), which triggers a card UI widget to pop up. Users follow commands on the user interface to navigate through the various stages involved in creating a music diary. The user initiates the process by clicking on an empty card (in Figure 3(a)) and then follows prompts for text inputs, during which the system displays ChatGPT responses (in Figure 3(b)). After the conversation ends, the system automatically uploads the results to the server to begin



Figure 3: The process of logging a diary entry

the creation of a diary entry, during which an animation indicates that the system is working (in Figure 3(c)). Finally, the system packages the diary entry (comprising text and corresponding emotion) for presentation to the user, during which a typing animation is presented and the soundtrack is played in the background (in Figure 3(d)).

2.2 ChatGPT Assistant

The proposed Music Diary utilizes the GPT-4-turbo-preview model and the ChatGPT Assistant API to generate prompts with three distinct functions. The first function involves querying the subject, wherein the GPT model operates as a good listener in conversational interactions to facilitate the exploration of various topics aimed at enhancing emotional awareness. Note that the system is configured to conclude conversations within 15 sentences. The second function involves summarizing first-person diary entries with the text content limited to fewer than 200 words. The third function involves identifying emotions in text for use by the music generation model. Note that the system is limited to the main emotion associated with a given entry.

2.3 Music Generation Model

This study employed the natural language-based music generation model developed by Audiogen [3], which is ideally suited to the task of generating music according to specific emotions. Note that Audiogen offers a stable and free API service for academic research. Note that this service allows the simultaneous generation of four 10-second music clips of reasonably good-quality. The user is given

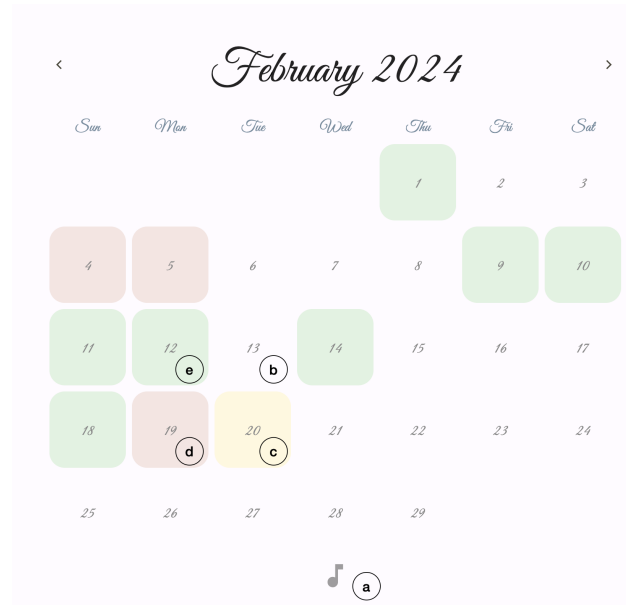


Figure 4: Diary entries are presented in a calendar view

the opportunity to select one of the four clips for inclusion in the diary entry. This is meant to empower the user to select the most appropriate music for a given emotional state.

3 PILOT STUDY

We recruited four volunteers (average age of 23.5 and equal gender distribution) to assess the Music Diary system. Participants were tasked with creating music diary entries for the current (experiment) day as well as the previous day. After the participants completed journaling process in the Music Diary app, they were asked to review each entry at least once and then fill out system usability surveys (SUS) [1] and other probing questions in a 5-level Likert scale pertaining to the user experience. Finally, we conducted brief semi-structured interviews to gather further insights.

The system consistently received good scores for system usability (mean score of 69.38 which is above average [1]), as indicated by the response to the questions *"The chatbot was helpful in completing the diary"* (score = 4) and *"The diary content generated by the chatbot is satisfactory,"* (score = 3.5).

The scores for the other probing questions were as follows: 1) the average score of *"The format of the music diary helped to enrich my emotional expression"* is 3.5; 2) the average score of *"The format of the music diary helped me to review past emotions"* is 4.0; 3) the average score of *"I am willing to share all or a part of the music diary content"* is 5.0. ; and 4) the average score of *"The sharing of music diary content could help others to understand my emotions"* is 4.0. These results indicate that Music Diary may enhance the willingness of users to share their emotions. Some of the participants reported that generating music in real-time felt like a creative process. They also mentioned that sharing the music with a few snippets of text would be an interesting endeavor.

4 CONCLUSION AND FUTURE WORKS

This paper presents a novel approach to enhancing retrospection and facilitating the creation of diary entries through the use of ChatGPT in conjunction with a music generation model. The proposed Music Diary system proved highly effective in promoting the writing of diary entries and exploring emotional expression, emotional awareness, and emotional reflection. In the next phase, we will expand the app to run on mobile devices and explore how to optimize the user interface for a more intuitive and user-friendly experience. Additionally, our focus will include creating a safe space where users feel comfortable sharing music clips and diary content. By encouraging sharing and positive interactions, we aim to provide more incentives for users to develop a journaling habit, ultimately promoting mental and physical well-being.

ACKNOWLEDGMENTS

This research was supported by the National Science and Technology Council of Taiwan (NSTC 113-2221-E-007-110 and 112-2221-E-007-095).

REFERENCES

- [1] 2011. SUSstified? Little-Known System Usability Scale Facts. <https://uxpamagazine.org/sustified/>. [Accessed 19-02-2024].
- [2] 2024. Audio Journal, Voice Diary, Lower Burnout - Journify — journify.co. [Accessed 19-02-2024].
- [3] 2024. Audiogen - Powering audio creation with generative AI — [audiogen.co](https://www.audiogen.co). [Accessed 19-02-2024].
- [4] 2024. Clarity - CBT Thought Diary — cbthoughtdiary.com. [Accessed 19-02-2024].
- [5] 2024. Emotion Journal: Track your Moods — Improvus — [teamimprovus.net](https://www.teamimprovus.net/emotion-journal). [Accessed 19-02-2024].
- [6] 2024. Journal — [apps.apple.com](https://apps.apple.com/us/app/journal/id6447391597). [Accessed 19-02-2024].
- [7] 2024. Mojitto - Daily Emoji Diary — [apps.apple.com](https://apps.apple.com/us/app/mojitto-daily-emoji-diary/id1631717082). [Accessed 19-02-2024].
- [8] 2024. The simplest way to get started with keeping a diary | Grid Diary — griddiaryapp.com. [Accessed 19-02-2024].
- [9] Zohar Elyoseph, Dorit Hadar-Shoval, Kfir Asraf, and Maya Lvovsky. 2023. ChatGPT outperforms humans in emotional awareness evaluations. *Frontiers in Psychology* 14 (2023), 1199058.
- [10] CT Rodriguez Hidalgo, Eduard S-H Tan, and Peeter WJ Verlegh. 2015. The social sharing of emotion (SSE) in online social networks: A case study in Live Journal. *Computers in Human Behavior* 52 (2015), 364–372.
- [11] Roger Hiemstra et al. 2001. Uses and benefits of journal writing. *New directions for adult and continuing education* 2001, 90 (2001), 19.
- [12] Bernard Rime, Batja Mesquita, Stefano Boca, and Pierre Philippot. 1991. Beyond the emotional event: Six studies on the social sharing of emotion. *Cognition & Emotion* 5, 5-6 (1991), 435–465.
- [13] Sakamoto Ryuichi. 17-01-2023. 12. online stream services.